

5.2 Energy: GPE & KE

Question Paper

Course	CIE A Level Physics
Section	5. Work, Energy & Power
Topic	5.2 Energy: GPE & KE
Difficulty	Medium

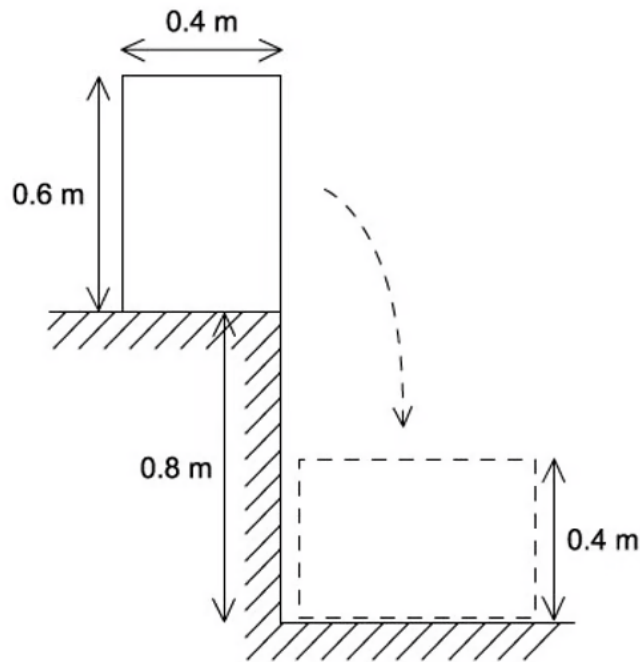
Time allowed: 20

Score: /9

Percentage: /100

Question 1

A uniform solid block has weight 500 N, width 0.4 m and height 0.6 m. The block rests on the edge of a step of depth 0.8 m, as shown.



The block is knocked over the edge of the step and rotates through 90° before coming to rest with the 0.6 m edge horizontal.

What is the change in gravitational potential energy of the block?

- A. 300 J
- B. 400 J
- C. 450 J
- D. 550 J

[1 mark]

Question 2

A body travelling with a speed of 20 m s^{-1} has kinetic energy E_k

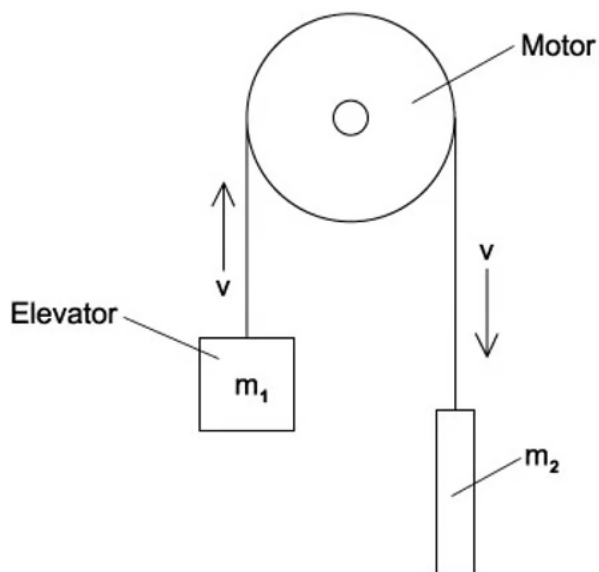
If the speed of the body is increased to 80 m s^{-1} what is its kinetic energy?

- A. $4E_k$
- B. $8E_k$
- C. $12E_k$
- D. $16E_k$

[1 mark]

Question 3

The diagram shows a lift system in which the elevator (mass m_1) is partly counterbalanced by a heavy weight (mass m_2)



At what rate does the motor provide energy to the system when the elevator is rising at a steady speed v ? (g = acceleration of free fall)

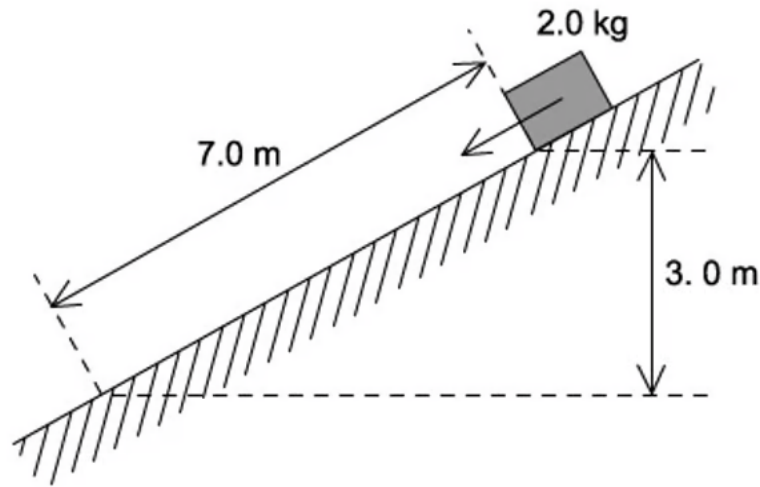
- A. $\frac{1}{2} m_1 v^2$
- B. $\frac{1}{2} (m_1 - m_2) v^2$
- C. $m_1 g v$
- D. $(m_1 - m_2) g v$

[1 mark]

Question 4

A block of mass 2.0 kg is released from rest on a slope. It travels 7.0 m down the slope and falls a vertical distance of 3.0 m

The block experiences a frictional force parallel to the slope of 5.0 N .



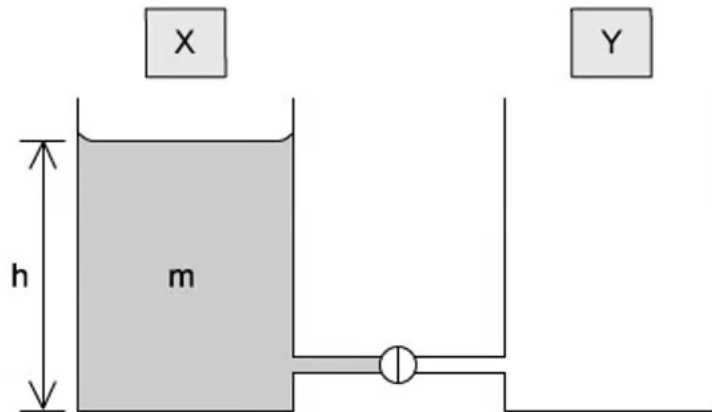
What is the speed of the block after falling this distance?

- A. 4.9 m s^{-1}
- B. 6.6 m s^{-1}
- C. 8.6 m s^{-1}
- D. 10.1 m s^{-1}

[1 mark]

Question 5

The diagram shows two identical vessels X and Y connected by a short pipe with a tap.



Initially, X is filled with water of mass m to a depth h , and Y is empty.

When the tap is opened, water flows from X to Y until the depths of water in both vessels are equal.

How much potential energy is lost by the water during this process? (g = acceleration of free fall)

- A. 0
- B. $\frac{mgh}{4}$
- C. $\frac{mgh}{2}$
- D. mgh

[1 mark]

Question 6

A concrete cube of side 0.60 m and uniform density $2.0 \times 10^3 \text{ kg m}^{-3}$ is lifted 5.0 m vertically by a crane.

What is the change in potential energy of the cube?

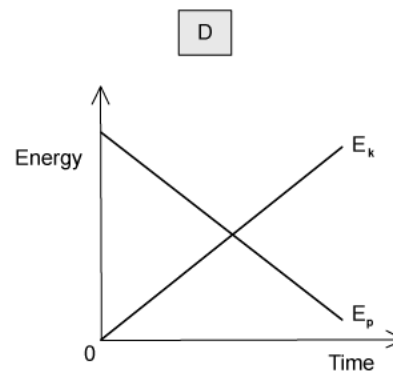
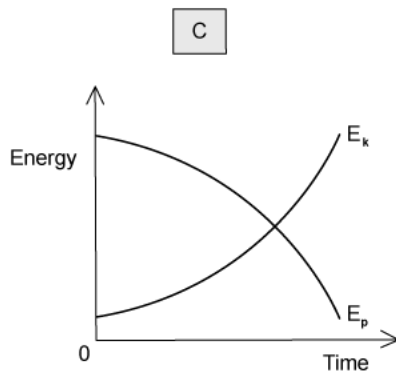
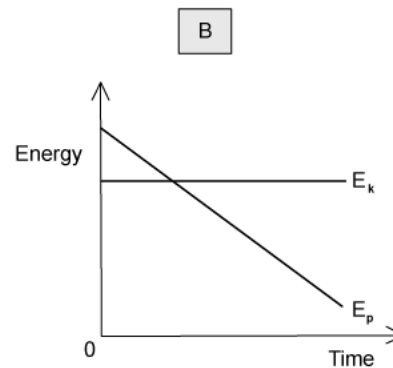
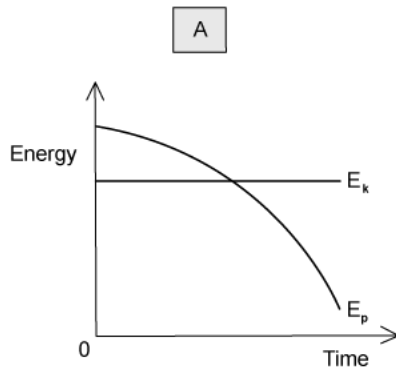
- A. 0.75 kJ
- B. 7.4 kJ
- C. 21 kJ
- D. 470 kJ

[1 mark]

Question 7

A steel ball is falling at constant speed in oil.

Which graph shows the variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball?



[1 mark]

Question 8

Car X is travelling at half the speed of car Y. Car X has twice the mass of car Y.

Which statement is correct?

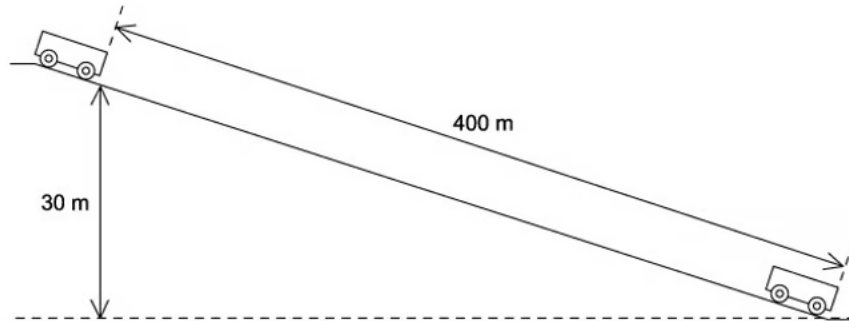
- A. car X has half the kinetic energy of car Y
- B. car X has one quarter of the kinetic energy of car Y
- C. car X has twice the kinetic energy of car Y
- D. the two cars have the same kinetic energy

[1 mark]

Question 9

A truck of mass 500 kg moves from rest at the top of a section of track 400 m long and 30 m high, as shown.

The frictional force acting on the truck is 250 N throughout its journey



What is the final speed of the truck?

- A. 14 m s^{-1}
- B. 24 m s^{-1}
- C. 31 m s^{-1}
- D. 190 m s^{-1}

[1 mark]

The correct answer is **D** because:

- The rate at which the motor provides energy is the power of the motor
- In the lift system, the mass m_2 works downwards under gravity, causing the elevator of mass m_1 to move up. However, the motor also provides some energy in moving the elevator up
- The gravitational potential energy of the masses is changing continuously and at a steady rate, since they are moving at a steady speed
- GPE of elevator = m_1gh_1
 - Where h_1 is its height above the ground
- Rate of change of GPE of elevator = $\frac{m_1gh_1}{t} = m_1gv$
- GPE of heavy weight = m_2gh_2
 - Where h_2 is its height above the ground
- Rate of change of GPE of heavy weight = $\frac{m_2gh_2}{t} = m_2gv$
- Let the power of the motor be P
 - $m_1gv = m_2gv + P$
 - $P = (m_1 - m_2)gv$
- Therefore, the power of the motor is equal to $(m_1 - m_2)gv$

A & B are incorrect because these choices have units of energy, not power. If **B** was correct, then there would have no need for the heavy weight

C is incorrect because this option ignores the influence of the other mass m_2

4

The correct answer is **A** because:

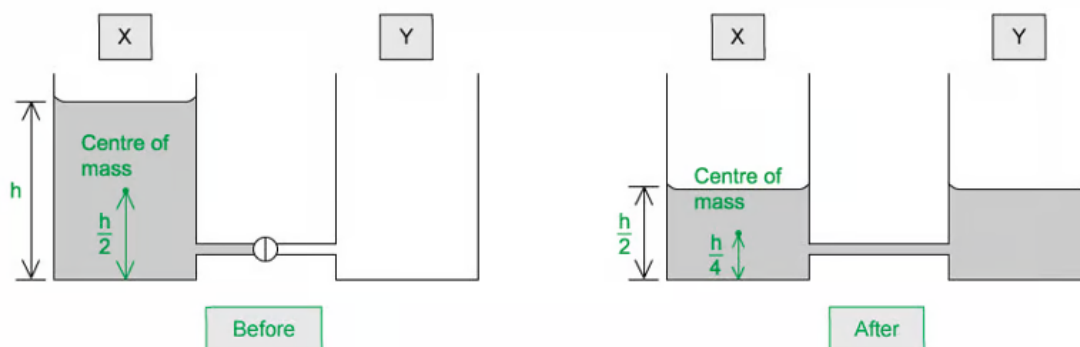
- At the top of the slope, the block has gravitational potential energy (GPE). As the block falls, some GPE is converted into kinetic energy of the block and work done due the frictional force
- $\Delta GPE = KE + \text{Work done against friction}$
- The loss in height Δh of the block is 3.0 m
 - $\Delta GPE = mg\Delta h = 2.0 \times 9.81 \times 3.0 = 58.86 \text{ N}$
- Frictional force = 5.0 N

- Distance moved against friction = 7.0 m
 - Work done against friction $F \times d = 5.0 \times 7.0 = 35.0J$
- $\Delta GPE = KE + \text{Work done against friction}$
 - $58.86 = KE + 35.0$
- KE of block at bottom = $58.86 - 35.0 = 23.86J$
 - $\frac{1}{2}mv^2 = 23.86$
 - $v = \sqrt{\frac{23.86 \times 2}{2.0}} = 4.88 \text{ m s}^{-1}$
- Therefore, the speed of the block after falling is 4.9 m s^{-1}

5

The correct answer is **B** because:

- Consider the water in both vessels as one body with one mass
- Since the mass of the water is spread all over the volume it occupies, the centre of mass should be considered when calculating the gravitational potential energy (GPE)
- Before the tap is opened, the centre of mass of the water in vessel X may be considered to be at its centre which is at half the water level $\frac{h}{2}$
 - Potential energy of water before tap is opened = $mg\left(\frac{h}{2}\right) = \frac{mgh}{2}$
- When the tap is opened the total mass of water in the whole body is the same, but the water level is now at a height of $\frac{h}{2}$ in both vessels
- The centre of mass may be considered to be at the centre of the volume of water, which is, again, at a height of half the water level so the centre of gravity is now at a height of $\frac{h}{4}$
 - Potential energy of water after tap has been opened = $mg\left(\frac{h}{4}\right) = \frac{mgh}{4}$
- Loss in potential energy $\left(\frac{mgh}{2}\right) - \left(\frac{mgh}{4}\right) = \frac{mgh}{4}$



6

The correct answer is **C** because:

- The density of the cube is given as $\rho = 2.0 \times 10^3 \text{ kg m}^{-3}$
- The volume of the cube is given as $V = a^3 = (0.6)^3$
- Density is equal to $\rho = \frac{m}{v}$
 - So, mass is equal to $m = \rho \times V = 2.0 \times 10^3 \times (0.6)^3 = 432 \text{ kg}$
- Now, we have to find the change in potential energy of the cube. At ground level, the potential energy is equal to zero and the potential energy at height h is given by
 - $GPE = mgh = 432 \times 9.81 \times 5.0 = 21\,190 \text{ J}$
- Therefore, the change in GPE is 21 kJ

7

The correct answer is **B** because:

- Kinetic energy (KE) is equal to $E_k = \frac{1}{2}mv^2$
- As the speed is constant, KE also remains constant
 - On the graph, this will be shown as a straight, horizontal line
- Gravitational potential energy (GPE) is equal to $E_p = mgh$
- As the speed is constant, the rate at which the distance h decreases is also constant, so the decrease in GPE is also constant

- On the graph, this will be shown as a straight, diagonal line starting from a maximum value with a negative gradient

A & C are incorrect because the speed, and hence the change in distance, of the steel ball is constant, meaning the relationship between these quantities and E_k and E_p will be linear which would not result in curved lines on the graph

D is incorrect because this graph would be accurate for an object accelerating towards the ground, however the ball is travelling at a constant speed so the KE must be constant

8

The correct answer is **A** because:

- Let the speed of car X = v
- Let the mass of car X = $2m$
- Let the speed of car Y = $2v$
- Let the mass of car Y = m
 - Kinetic energy (KE) is equal to $E_k = \frac{1}{2}mv^2$
- KE of car X = $\frac{1}{2}(2m)v^2 = mv^2$
- KE of car Y = $\frac{1}{2}m(2v)^2 = 2mv^2$
 - Ratio $\frac{KE \text{ of car X}}{KE \text{ of car Y}} = \frac{mv^2}{2mv^2} = \frac{1}{2}$
- Therefore, car X has half the kinetic energy of car Y

9

The correct answer is **A** because:

- From the conservation of energy:
 - GPE of truck at top = KE of truck at bottom + Work done against friction
- GPE of truck at top = $mgh = 500 \times 9.81 \times 30 = 147\,150J$
- Work done against friction = $F \times d = 250 \times 400 = 100\,000J$
- KE of truck at bottom = GPE of truck at top – Work done against friction

$$\circ \quad \frac{1}{2}mv^2 = 147\,150 - 100\,000 = 47\,150$$
$$\sqrt{2 \times 47\,150}$$